Automatic Control Systems: Open-Loop vs. Closed-Loop

Why we close the loop: disturbance rejection and reduced sensitivity.

Ogata, *Modern Control Engineering*, Ch. 1–2.

In this tutorial you will:

- compare open-loop and closed-loop step responses,
- compute the steady-state error reduction from feedback, and
- see feedback reject an input disturbance.

Step through with **Ctrl+Enter**, or render a report with `publish`.

An **open-loop** control system uses a controller and process (plant) in a forward path only – the output is not measured or compared to the reference, so accuracy depends entirely on calibration and is not corrected for disturbances:

$Y(s) = G_c(s)G_p(s)R(s)$

A **closed-loop** (feedback) system measures the output, compares it to the reference through a summing junction, and drives the controller from the resulting error $E(s) = R(s) - H(s)Y(s)$:

$Y(s) = \frac{G_c(s)G_p(s)}{1+G_c(s)G_p(s)H(s)}R(s)$

Feedback trades some open-loop simplicity for reduced sensitivity to plant variations and disturbances, at the cost of potential instability if not designed carefully.

Contents

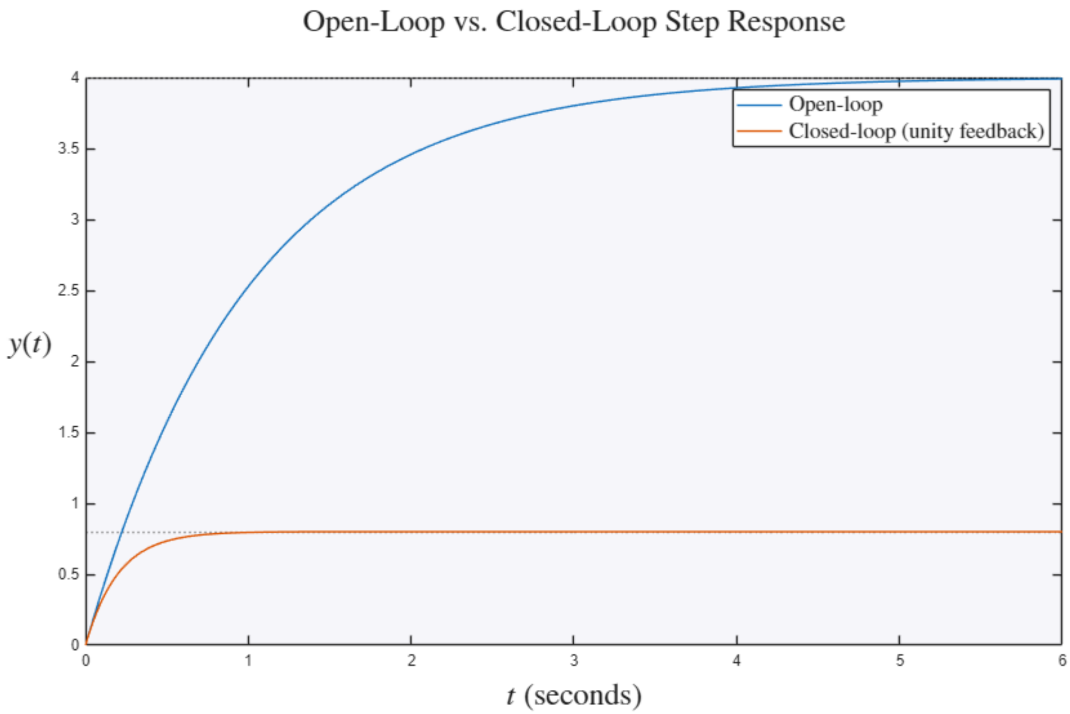
Plant and Controller

Consider a plant $G_p(s) = \frac{1}{s+1}$ controlled by a simple proportional controller $G_c(s) = K$, with unity feedback $H(s) = 1$.

```
Gp = tf(1,[1 1]);
K = 4;
Gc = K;

T_open = series(Gc,Gp);          % open-loop output, no feedback
T_closed = feedback(series(Gc,Gp),1);

figure
hold on
step(T_open)
step(T_closed)
hold off
legend('Open-loop','Closed-loop (unity feedback)','Interpreter','latex','FontSize',14)
title('Open-Loop vs. Closed-Loop Step Response','Interpreter','latex','FontSize',20)
ylabel('$y(t)$','Interpreter','latex','FontSize',20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$','Interpreter','latex','FontSize',20)
```



Note the open-loop system does not track $r = 1$: its unit-step output settles to $KG_p(0) = 4$, with nothing to correct it toward the reference. The closed-loop system's steady-state error is instead reduced by the loop gain:

$e_{ss} = \frac{1}{1+KG_p(0)H(0)} = \frac{1}{1+4} = 0.2$

```
ess_closed = 1/(1+K*dcgain(Gp))
```

```
ess_closed =

    0.2000
```

Disturbance Rejection

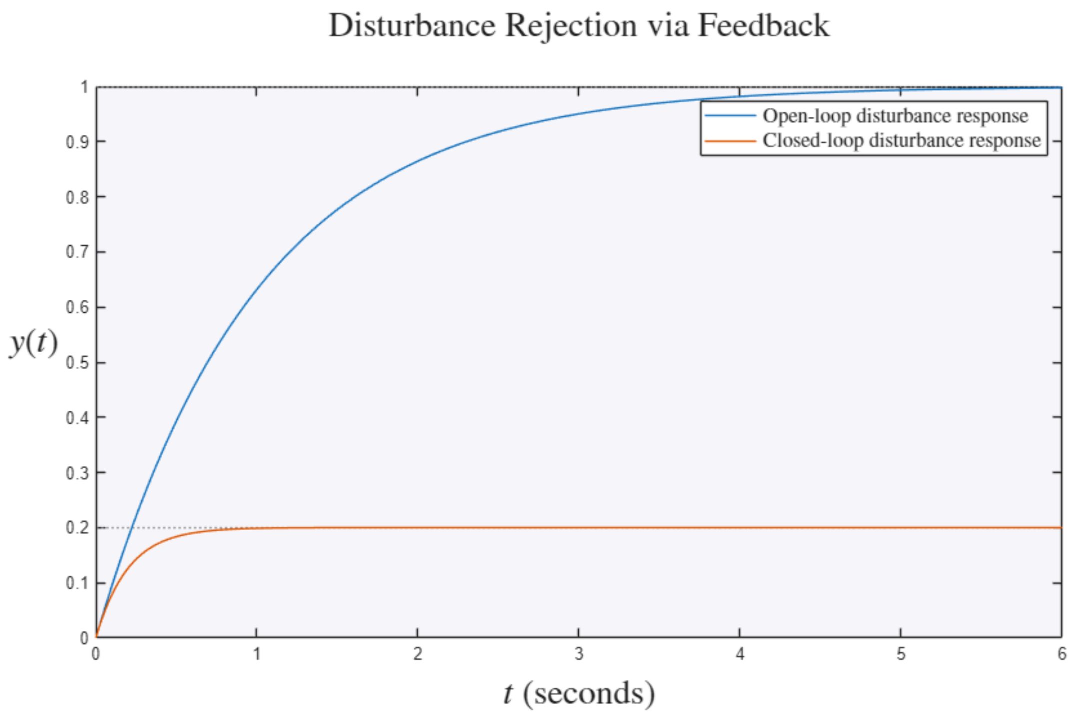
A key advantage of feedback is disturbance rejection. Add a disturbance $D(s)$ entering at the plant input:

$Y(s) = \frac{G_c G_p}{1+G_c G_p H} R(s) + \frac{G_p}{1+G_c G_p H} D(s)$

As the loop gain $G_c G_p H$ grows, the disturbance's contribution to $Y(s)$ shrinks, even though G_p itself is unchanged.

```
T_dist_open = Gp;                          % open-loop: D feeds Y directly through Gp
T_dist_closed = feedback(Gp,Gc);           % closed-loop: D path attenuated by loop gain

figure
hold on
step(T_dist_open)
step(T_dist_closed)
hold off
legend('Open-loop disturbance response','Closed-loop disturbance response','Interpreter','latex','FontSize',12)
title('Disturbance Rejection via Feedback','Interpreter','latex','FontSize',20)
ylabel('$y(t)$','Interpreter','latex','FontSize',20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$','Interpreter','latex','FontSize',20)
```



Try it yourself

- Raise `K` to 20 and notice the closed-loop steady-state error 1/(1+K) shrink further while the open loop drifts even farther off.
- Halve the plant gain and watch the closed loop barely move – feedback desensitizes the output to plant changes.